

Expression, Purification, and Characterization of Recombinant Human Factor X¹

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A system is described for producing recombinant factor X with properties very similar to human plasma factor X. Optimization of the expression system for factor X resulted in the finding that human kidney cells (293 cells) are superior to the widely utilized baby hamster kidney cells (BHK cells) for the expression of functional factor X. It was also determined that production of factor X by 293 cells requires the substitution of the –2 residue (Thr → Arg) which affords the removal of the factor X propeptide. Purification of recombinant and plasma factor X is accomplished using a calcium-dependent monoclonal antibody directed against the gla domain. The proteins are comparable by sodium dodecyl sulfate polyacrylamide gel electrophoresis. The rate and extent of activation by the factor X coagulant protein from Russell's viper venom and by factors IXa and VIIIa are similar; activation of the recombinant protein by VIIa and tissue factor is mildly faster. The activated enzymes have the same activity toward a chromogenic substrate and the biologic substrate, prothrombin. Both enzymes have the same apparent affinity for the activated platelet surface as judged by their ability to activate prothrombin. Finally, inhibition by antithrombin, with or without heparin, and inhibition by the tissue factor pathway inhibitor are equivalent. Recombinant factor X produced by this method is therefore well suited for probing structure–function relationships by mutational analysis. © 1997 Academic Press

The vitamin K-dependent plasma serine protease factor Xa holds a pivotal position in blood coagulation as the only known physiological activator of prothrom-

bin. Factor X activates prothrombin efficiently on a phospholipid surface in the presence of a cofactor, factor Va (1). The zymogen precursor factor X is activated to factor Xa in the presence of calcium by complexes in the intrinsic (fIXa/fVIIIa)² and extrinsic (fVIIa/TF) pathways and RVV-X (2–7). Activated factor X is inhibited by antithrombin III (ATIII) and tissue factor pathway inhibitor (TFPI, 8–11). The light chain of factor X (M_r 17,500) contains an amino-terminal gla domain, which includes 11 γ -carboxylated glutamic acid residues necessary for the calcium-dependent membrane-binding capacity of the molecule, and two epidermal growth factor-like domains (EGF-1 and EGF-2). The heavy chain (M_r 45,000) is joined to the light chain by a single disulfide bond and contains the activation peptide and the serine protease domain.

Better insight into its interactions with its activators, cofactor, substrates, and inhibitors is needed to improve understanding of this complex pathway. Expression of molecules with specific mutations using recombinant technology is a powerful method for correlating structure and function. But since protein processing can vary in different expression systems, the wild-type recombinant protein must be thoroughly characterized to ensure that its functional properties closely mirror those of the plasma protein.

Human kidney cells (293 cells; ATCC, CRL 1573) have been used to express human factor X (12–14). Herein we report for the first time the characterization of recombinant human factor X and a detailed comparison of factor X expressed in 293 cells with plasma-derived factor X.

² Abbreviations used: TF, tissue factor; fVIIa, fVa, fIXa, fXa, and fVIIIa, activated factors VII, V, IX, X, and VIII; RVV-X, factor X-activating protein from Russell's viper venom; PC, phosphatidylcholine; PS, phosphatidylserine; FPLC, fast protein liquid chromatography; MES, 4-morpholineethanesulfonic acid; ATIII, antithrombin III; TFPI, tissue factor pathway inhibitor.

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Factor X secreted by transfected 293 cells and purified with a calcium-dependent monoclonal antibody has functional properties nearly indistinguishable from factor X purified from plasma. This expression system will be useful for producing specific factor X variants to correlate structure and function.

MATERIALS AND METHODS

Reagents and chemicals. Spectrozyme TH (*H*-D-hexahydrotyrosyl-L-alanyl-L-arginine-*p*-nitroanilide diacetate) and Spectrozyme FXa (methoxycarbonyl-D-hexahydrotyrosyl-L-alanyl-L-arginine-*p*-nitroanilide diacetate) were obtained from American Diagnostica (Greenwich, CT). Factor X-deficient plasma was purchased from George King Bio-Medicals, Inc. (Overland Park, KS). All other reagents and chemicals were of highest quality available commercially.

Proteins. Three murine monoclonal antibodies used for purification and immunoassay have been previously described (15). Prothrombin was purified from human plasma, concentrated (to >1 mM), and immunodepleted of residual factor X (<1 nM final) by passage through a column of immobilized antibody 3448.1D7.20 as previously described (15). ATIII was purchased from Kabi Pharmacia Diagnostics (Piscataway, NJ). Recombinant TFPI was a gift from Monsanto/Searle Company (St. Louis, MO). Factors Va, VIIa, and IXa from human plasma were purchased from Haematologic Technologies (Essex Junction, VT). Thrombin was prepared by cation-exchange chromatography after activation of human prothrombin with Taipan snake (*Oxyuranus scutellatus*) venom. Porcine factor VIII was obtained from Porton Products (Agoura Hill, CA) and was further purified (16) using an immobilized antibody (W-3, a generous gift from Dr. David Fass). Innovin, a lipitated, recombinant tissue factor, was obtained from Baxter Diagnostics (Deerfield, IL). RVV-X was purified from Russell's viper venom (17).

Plasma and recombinant factor X were purified as described (15). Briefly, conditioned medium was collected daily from 3×10^8 cells for 7 days and frozen. Medium was thawed at 37°C, adjusted to 32 mM citrate, and 1 M barium chloride was added dropwise to a final concentration of 70 mM. The calcium-binding properties of the gla domain afford the adsorption of factor X to the barium citrate precipitate. Following barium citrate adsorption and elution with 20% saturated $(\text{NH}_4)_2\text{SO}_4$ containing 5 mM diisopropylfluorophosphate, the resulting supernatant was collected by centrifugation at 10,000*g* and dialyzed overnight against 100 mM NaCl, 1 mM benzamidine, 10 mM Hepes, pH 7.0. The solution was clarified by centrifugation at 12,000*g* and concentrated by ultrafiltration using an Amicon flow cell with a YM30 membrane until the factor X concentration was at least 2 μM . After addition

of 1.0 M CaCl_2 to bring the final concentration to 5 mM, the solution was loaded at 4°C onto a column of Affigel 10 (Bio-Rad, Richmond, CA) to which a murine monoclonal antibody 3698.1A8.10 had been coupled (2 mg/ml of gel) according to the manufacturer's instructions. As described previously, 3698.1A8.10 only binds factor X in the presence of calcium ions, as it is directed against the gla domain (15). Separate columns were used for plasma and recombinant factor X. The column was washed with a buffer containing 100 mM NaCl, 5 mM CaCl_2 , 10 mM Hepes, pH 7.0, until the absorbance at 280 nm was less than 0.01. The factor X was then eluted with the same buffer containing 5 mM EDTA instead of CaCl_2 . Fractions containing factor X were pooled and loaded onto a Pharmacia Mono Q FPLC column. The factor X was eluted with 500 mM NaCl, 10 mM Hepes, pH 7.0, and then adjusted to 2.5 mM CaCl_2 and pH 6.5 (with MES), sterile filtered, made 20% in sterile glycerol, aliquotted, snap frozen in liquid nitrogen, and stored at -70°C. Protein concentration was determined by absorbance at 280 nm using an extinction coefficient of 11.6 (1%, 1 cm) (2), and by the two-site immunoassay described below using factor X purified from plasma as the standard.

Transfection and cell culture. Construction, transfection, and expression of wild-type factor X in 293 cells has been previously described (15). Briefly, a full-length factor X cDNA was cloned from a human hepatocyte library. The codon corresponding to the -2 residue of factor X was changed (ACG → AGG, Thr → Arg) to allow correct cleavage of the propeptide. The entire wild-type cDNA sequence was verified. The ZMB3 expression vector was a gift from Dr. Don Foster (Zymogenetics). Human kidney cells (293 cells; ATCC, CRL 1573) were transfected by calcium phosphate precipitation (18) and cultured as described previously (15). Isolated colonies were screened for factor X production by immunoassay before expansion of selected clones for medium conditioning.

Modified prothrombin time assay. Samples were incubated for 5 min at 37°C with factor X-deficient plasma. The clotting time was measured using a fibrometer (BBL Microbiology Systems, Cockeysville, MD) following addition of thromboplastin (Ortho, Raritan, NJ).

Inhibition studies. Factor Xa (0.5 nM) was incubated with 100 μM Spectrozyme FXa and varying concentrations of ATIII (0–4 μM), TFPI (0–50 nM), or ATIII (0–16 nM) in the presence of subsaturating heparin (2.5 mU/ml). Progress curves representing residual factor Xa activity were fitted, using the Marquardt-Levy algorithm (SigmaPlot; Jandel Scientific, Corte Madera, CA), to an exponential equation of the form

$$A = A_f(1 - e^{-k_{\text{obs}}t}) + A_0,$$

where A is the observed absorbance, A_f is the amplitude of the curve, k_{obs} is the apparent first-order rate constant (units of s^{-1}), t is time in seconds, and A_0 is the initial absorbance. Each derived apparent rate constant (k_{obs}) was then plotted as a linear function of the effective inhibitor concentration and the resulting slope calculated to estimate second-order rate constants.

Other. Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS/PAGE) was performed by the method of Laemmli (19). Proteins were stained directly using Coomassie blue. Egg yolk phosphatidylcholine (PC) and bovine brain phosphatidylserine (PS) were purchased from Sigma (St. Louis, MO). PC:PS vesicles (3:1 ratio) of nominal 100-nm diameter were synthesized by the method of membrane extrusion (20) in 10 mM Hepes, pH 7.5, 100 mM NaCl, 5 mM CaCl_2 , at a total lipid concentration of 1.0 mM using the Lipofast Basic device (Avestin, Ottawa, Ontario, Canada) and were then made 10% in glycerol, frozen in liquid nitrogen, and stored at -70°C .

RESULTS AND DISCUSSION

Expression

Initially, a number of stable, cloned lines that expressed factor X with $<1\%$ activity were established. The problem, documented by protein sequencing of the light chains of recombinant protein purified from conditioned media, was failure to remove the entire propeptide from the secreted protein. The two major products retained 7 and 15 amino acids. The enzyme responsible for cleaving the propeptide from this and other molecules has been identified (21,22) and termed paired basic amino acid-cleaving enzyme. Cleavage is most efficient when the -1 residue is Arg, the -2 residue is Lys or Arg, and the -4 residue is Arg. Therefore, the codon corresponding to the -2 residue of wild-type factor X was mutated (Thr $\rightarrow$ Arg), and expression of functional recombinant factor X was achieved.

Baby hamster kidney cells (BHK cells) were utilized initially for expression of factor X because this cell line is known to efficiently express a homologous protein, coagulation factor VII (23,24). Based on clotting activity, the specific activity of BHK factor Xa in media is approximately half that of plasma factor Xa (data not shown). It was determined that BHK factor X demonstrates reduced reactivity with a calcium-dependent monoclonal antibody directed against the gla domain, suggesting that γ -carboxylation of BHK factor X is incomplete. Factor X expressed in human kidney cells, 293 cells, however, is fully functional compared to plasma factor X prior to purification in a plasma-based assay (see below).

Purification

The concentration of recombinant factor X varied from 0.4 to 1.0 $\mu\text{g}/\text{ml}$ (7 to 20 nM) in media conditioned

for 24 h on confluent cells compared with plasma factor X, 8 $\mu\text{g}/\text{ml}$ (140 nM). The final yield of purified protein varied from 40 to 60% of the initial starting material (Table 1). The isolation scheme effects roughly a 5000- to 10,000-fold purification and 150- to 1600-fold concentration. A Coomassie-stained 12.5% SDS polyacrylamide gel comparing plasma to recombinant factor X is shown in Fig. 1. The heavy chain of the recombinant material migrates slightly slower. Once activated the enzymes are indistinguishable by SDS/PAGE (data not shown), indicating that the small difference in apparent mass is localized to the activation peptide. This difference is most likely the result of minor glycosylation variations since the cDNA sequence directing the correct wild-type amino acid sequence was confirmed on two separate occasions. The minor difference in apparent mass of the activation peptide does not significantly affect the function of the recombinant molecule (see below).

Plasma-Based Coagulant Activity

The concentration of purified plasma and recombinant factor X was calculated from their absorbance at 280 nm. Dilutions of each protein based on these concentrations show identical responses in the two-site monoclonal antibody-based immunoassay (data not shown). Recombinant factor X was found to be functionally equivalent to plasma factor X based on its ability to correct the clotting time of factor X-deficient plasma in a prothrombin time-based assay (data not shown). Since this global measure of function could be insensitive to particular aspects of activation, catalysis, or inhibition, these functions were individually tested using purified components.

Activation

Plasma or recombinant factor X (100 nM) in assay buffer (100 mM NaCl, 5 mM CaCl_2 , 0.1% bovine serum albumin, 0.1% polyethylene glycol 8000, 10 mM Hepes, pH 7.5) was activated by (1) 62 pM RVV-X, (2) 2 nM factor IXa and 4 units/ml thrombin-activated porcine factor VIIIa on 20 μM PC:PS vesicles, or (3) 20 pM factor VIIa in the presence of 500 pM lipidated recombinant human TF. Activation was monitored by quenching samples in assay buffer with 5 mM EDTA substituted for CaCl_2 and then measuring the rate of hydrolysis of 100 μM Spectrozyme FXa in a Molecular Devices kinetic microplate reader.

The rate and the extent of activation by limiting amounts of RVV-X are the same for plasma and recombinant factor X (Fig. 2A). Activations of plasma versus recombinant factor X by factors IXa and VIIIa are also very similar (data not shown). Recombinant factor X is activated somewhat faster than plasma factor X by factor VIIa and TF, as shown in Fig. 2B, with the maximal

TABLE 1
Purification of Recombinant Human Factor X

Purification step	Volume (ml) ^a	Factor X (μM) ^b	Recovery (%)	Yield (nmol) ^b
Media	3200	0.011	100	36.0
Barium adsorption	180	0.185	92.6	33.4
Ultrafiltration	6.10	5.03	85.0	30.7
Immunoaffinity	2.01	10.6	59.1	21.3
Ion exchange	1.02	17.6	50.0	18.0

^a 3×10^8 cells grown in monolayer were used to express ~450 ml medium/day. Conditioned medium was collected daily for 7 days.

^b Factor X M_r ~57,500.

turnover rate at saturating factor X estimated as 4.2 and 2.9 mol fXa/s/mol fVIIa, respectively. The K_m is the same for both substrates (65 and 64 nM, respectively).

Catalysis

In the activation experiments described above, the concentration of factor Xa formed was determined from the initial rate of hydrolysis of a chromogenic substrate. This approach is valid because comparable concentrations of both enzymes, as determined both by absorbance at 280 nm and by immunoassay, react similarly toward the chromogenic substrate. The amidolytic activity was determined by incubating each factor Xa at 0.5 nM with increasing concentrations of Spectrozyme FXa (0–1 mM) (data not shown). Both the K_m and the turnover rate for Spectrozyme FXa were deter-

mined to be equivalent for plasma and recombinant factor X (98 versus 115 μM, respectively and 256 versus 233 mol *p*-nitroaniline/s/mol fXa, respectively). This demonstrates that the catalytic pockets of both proteins are quite similar.

Prothrombin is the best characterized protein substrate for Xa. The cofactor, factor Va, and a negatively charged phospholipid surface, both of which interact with factor Xa and with prothrombin, are required for optimal thrombin formation. In the experiment shown in Fig. 3, these interactions were probed. The surface was provided by 20 μM PC:PS vesicles. The prothrombin was initially present at 1.0 μM, and the enzymes were tested at 20 pM. At these concentrations, the amount of prothrombin cleaved by factor Xa in the absence of factor Va is negligible. The factor Va concentration was varied from 20 to 1000 pM. The recombinant and plasma factor Xa are indistinguishable with apparent K_d s for factor Va of 58 and 57 pM and rates for prothrombin activation at saturating factor Va of 23 and 22 mol thrombin/s/mol fXa. These are very similar to reported values (25).

PC:PS vesicles have been extensively used in studies of coagulation proteins but do not necessarily reflect the surfaces available *in vivo*. The ability of the enzymes to generate thrombin was therefore also compared on activated platelets, which supply a surface and factor Va. The rate of thrombin formation from 1.0 μM prothrombin by 43 pM factor Xa was determined in the presence of 10^8 /ml activated platelets (data not shown). This concentration of enzyme was chosen because it is close to the apparent K_d (50 pM) of plasma factor Xa for the high-affinity (factor Va-dependent) binding sites on activated platelets (26,27). Aliquots of the reaction were removed over time (0–3 min) and quenched into EDTA-containing buffer. Initial rates of thrombin formation (plasma fXa 24.6 nM/min, recombinant fXa 23.4 nM/min) were quantitated using Spectrozyme TH hydrolysis as described in the legend to Fig. 3. Recombinant and plasma factor Xa activate prothrombin identically under these conditions.

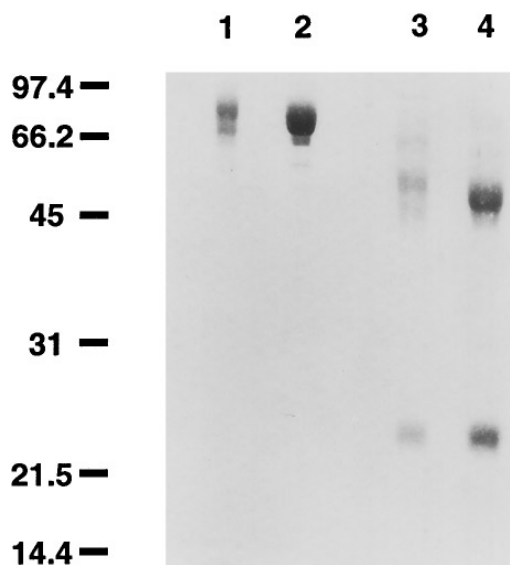


FIG. 1. Comparison of recombinant and plasma-derived factor X. SDS/polyacrylamide gel (12.5%) of nonreduced (lanes 1 and 2) and reduced (lanes 3 and 4) recombinant (lanes 1 and 3) or plasma (lanes 2 and 4) factor X.

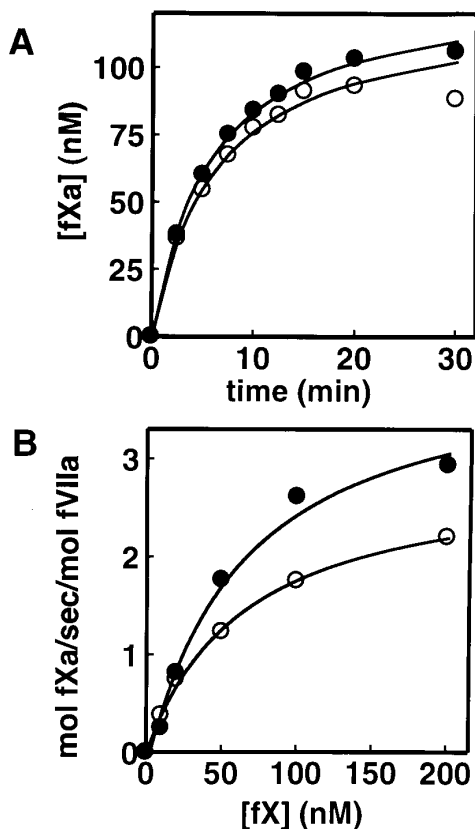


FIG. 2. Activation of plasma and recombinant factor X by RVV-X and TF/fVIIa complexes. (A) 62 pM RVV-X was incubated with 100 nM plasma (○) or recombinant (●) factor X. Samples were removed at the times indicated and quenched as described in the text, and the initial rate of Spectrozyme FXa (100 μ M) hydrolysis was determined. The concentration of factor Xa was calculated from a standard curve prepared from dilutions of maximally activated factor Xa. (B) Varying concentrations of plasma (○) or recombinant (●) factor X were activated with 20 pM fVIIa in the presence of 500 pM relipidated TF. The reactions were quenched after 5 min and the factor Xa concentrations were determined as described above.

Inhibition Studies

As a final test of function, the inhibition of plasma and recombinant factor Xa by ATIII and by TFPI were compared. Progress curves showing inhibition of the hydrolysis of 100 μ M Spectrozyme FXa by 0.5 nM factor Xa over 15 min by increasing concentrations (0–4 μ M) of antithrombin are presented in Fig. 4. Second-order rate constants were calculated as described under Materials and Methods ($1.52 \times 10^3 \text{ M}^{-1} \text{ s}^{-1}$ and $1.65 \times 10^3 \text{ M}^{-1} \text{ s}^{-1}$ for plasma and recombinant factor Xa, respectively). In additional experiments, the acceleration of inhibition by a subsaturating concentration (2.5 mU/ml) of full-length heparin was determined. ATIII concentrations were decreased to 0–16 nM to permit data collection over the same time interval (15 min). The derived rate

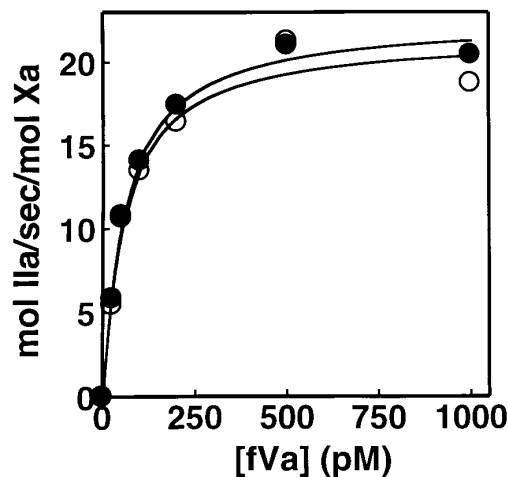


FIG. 3. Prothrombin activation by recombinant and plasma-derived factor X on PC:PS vesicles in the presence of fVa. Generation of thrombin from 1.0 μ M prothrombin by 20 pM plasma (○) or recombinant (●) factor Xa in the presence of 20 μ M PC:PS (3:1) phospholipid vesicles and the factor Va concentrations indicated. Exactly 5 min after the addition of prothrombin, samples were diluted 20-fold in EDTA-containing buffer and the initial rate of hydrolysis of 0.5 mM Spectrozyme TH was monitored at 405 nm. The concentration of thrombin was determined from a standard curve prepared from dilutions of maximally activated prothrombin. The rate of thrombin formation at each fVa concentration was then calculated and expressed as mol thrombin (IIa)/s/mol fXa.

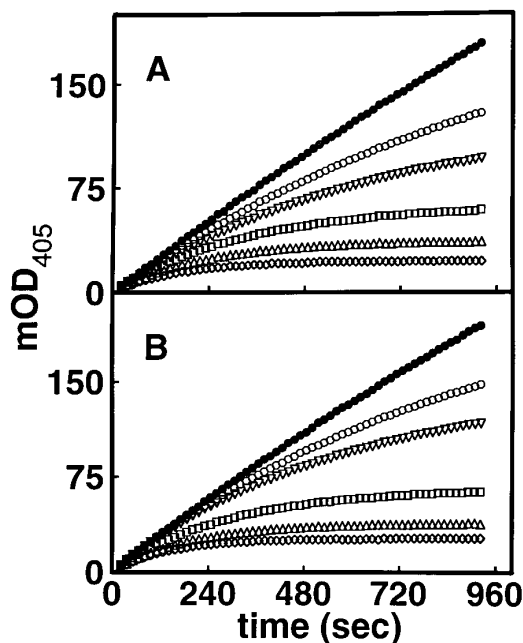


FIG. 4. Inhibition of recombinant and plasma-derived fXa. 0.5 nM plasma (A) or recombinant (B) factor Xa was inhibited by ATIII. The hydrolysis of 100 μ M Spectrozyme FXa was monitored for 15 min at 405 nm in the presence of 0 (●), 0.5 (○), 1.0 (▽), 2.0 (□), 3.0 (△), or 4.0 (◇) μ M ATIII.

constants were 1.17×10^6 and $1.12 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$ for plasma and recombinant factor Xa, respectively. Finally, the inhibition of 0.5 nM factor Xa by 0–50 nM TFPI was monitored in similar experiments. The estimated rate constants for inhibition were $1.34 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$ versus $1.33 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$ for plasma and recombinant factor Xa, respectively. These values are similar to previously reported values (28).

Taken together, these experiments demonstrate the functional equivalence of plasma and recombinant factor Xa in reactions that probe interactions with activators, substrates, the cofactor, and inhibitors. Recombinant factor X can be produced with properties that parallel plasma-derived factor X. Therefore, this expression system can be utilized for the expression of variant factor X for structural and functional studies.

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